Importance-Weighted Randomized Coordinate Descent for Non-Convex, Coordinate-Wise Smooth Functions

Algorithm Name

Importance-Weighted Randomized Coordinate Descent (IW-RCD).

Mathematical Formulation

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. Denote by $e_i \in \mathbb{R}^d$ the i-th standard basis vector and by $\nabla_i f(w)$ the i-th partial derivative at w. Let probabilities $p = (p_1, \ldots, p_d)$ satisfy $p_i > 0$ and $\sum_{i=1}^d p_i = 1$.

At iteration $k \in \{0, 1, 2, \ldots\}$: - Sample a coordinate $i_k \in \{1, \ldots, d\}$ with $\mathbb{P}(i_k = i) = p_i$. - Update only coordinate i_k according to

$$w^{k+1} \;=\; w^k \;-\; \eta_k \, \frac{1}{p_{i_k}} \, \nabla_{i_k} f(w^k) \, e_{i_k},$$

where $\eta_k > 0$ is a stepsize.

This can be equivalently viewed as a stochastic-gradient step with the unbiased estimator

$$g^k \;:=\; \frac{1}{p_{i_k}} \, \nabla_{i_k} f(w^k) \, e_{i_k}, \qquad \mathbb{E}\Big[g^k \mid w^k\Big] \;=\; \nabla f(w^k).$$

Stepsize constraint used in the analysis:

$$0 < \eta_k \;\leq\; \min_{1 \leq i \leq d} \frac{p_i}{L_i},$$

where $L_i > 0$ are coordinate-wise smoothness constants (defined below).

Pseudocode

Inputs: initial point $w^0 \in \mathbb{R}^d$; probabilities $p \in \Delta^{d-1}$ with $p_i > 0$; stepsizes $\{\eta_k\}_{k \geq 0}$; maximum iterations T or tolerance $\varepsilon > 0$.
For $k = 0, 1, 2, \ldots$ until $k = T - 1$ or $\left\|\nabla f(w^k)\right\| \leq \varepsilon$:

1. Sample $i_k \sim p$.
2. Compute the partial derivative $g_k := \nabla_{i_k} f(w^k)$.
3. Update: $w^{k+1} \leftarrow w^k - \eta_k \frac{1}{p_{i_k}} g_k \, e_{i_k}$.

Return w^T or the last iterate that satisfies the stopping criterion.

Dry Run Example

Consider $d = 1$ with $f(w) = (w-3)^2$, $w^0 = 0$, $\eta_k \equiv \eta = 0.1$, and $p_1 = 1$. - Gradient: $\nabla f(w) = 2(w-3)$. - Update: $w^{k+1} = w^k - \eta \cdot \nabla f(w^k) = w^k - 0.2\,(w^k - 3) = 0.8\,w^k + 0.6$.

Iterates and function values:

- $k = 0$: $w^0 = 0$, $\nabla f(w^0) = 2(0-3) = -6$, $w^1 = 0 - 0.1(-6) = 0.6$, $f(w^0) = 9$, $f(w^1) = (0.6-3)^2 = 5.76$.
- $k = 1$: $w^1 = 0.6$, $\nabla f(w^1) = 2(0.6-3) = -4.8$, $w^2 = 0.6 - 0.1(-4.8) = 1.08$, $f(w^2) = (1.08-3)^2 = 3.6864$.
- $k = 2$: $w^2 = 1.08$, $\nabla f(w^2) = 2(1.08-3) = -3.84$, $w^3 = 1.08 - 0.1(-3.84) = 1.464$, $f(w^3) = (1.464-3)^2 = 2.359296$.

Assumptions

Assumption 1 (Lower boundedness)**.** *There exists $f_\star > -\infty$ such that $f(w) \geq f_\star$ for all $w \in \mathbb{R}^d$.*

Assumption 2 (Coordinate-wise smoothness)**.** *For each coordinate $i \in \{1, \dots, d\}$, there exists $L_i > 0$ such that for all $w \in \mathbb{R}^d$ and $h \in \mathbb{R}$,*

$$|\nabla_i f(w + he_i) - \nabla_i f(w)| \leq L_i |h|.$$

Equivalently, for all $w \in \mathbb{R}^d$ and $h \in \mathbb{R}$,

$$f(w + he_i) \leq f(w) + \nabla_i f(w)\, h + \frac{L_i}{2} h^2.$$

Assumption 3 (Sampling distribution)**.** *Sampling probabilities $p = (p_1, \dots, p_d)$ satisfy $p_i > 0$ and $\sum_{i=1}^d p_i = 1$.*

Assumption 4 (Stepsizes)**.** *The stepsizes $\{\eta_k\}_{k \geq 0}$ satisfy*

$$0 < \eta_k \leq \min_{1 \leq i \leq d} \frac{p_i}{L_i} \qquad \text{for all } k \geq 0.$$

Main Convergence Theorem

Theorem 1 (Expected descent and nonconvex stationarity rate)**.** *Under the assumptions above, the IW-RCD iterates $\{w^k\}_{k \geq 0}$ satisfy, for every $T \geq 1$,*

$$\sum_{k=0}^{T-1} \frac{\eta_k}{2}\, \mathbb{E}\Big[\left\| \nabla f(w^k) \right\|^2 \Big] \;\leq\; \mathbb{E}[f(w^0)] - f_\star \;=\; f(w^0) - f_\star.$$

In particular, letting $\eta_{\min} := \min_{0 \leq k \leq T-1} \eta_k$ and choosing a random index R uniformly in $\{0, \dots, T-1\}$ independent of the algorithm's history,

$$\mathbb{E}\big[\|\nabla f(w^R)\|^2 \big] \;\leq\; \frac{2\,(f(w^0) - f_\star)}{\eta_{\min}\, T}.$$

If $\eta_k \equiv \eta \leq \min_i p_i / L_i$, then $\mathbb{E}[\left\|\nabla f(w^R)\right\|^2] = \mathcal{O}(1/T)$ with constant $2(f(w^0) - f_\star)/\eta$.

Theoretical Properties

- Monotone expected descent: $\mathbb{E}[f(w^{k+1})] \le \mathbb{E}[f(w^k)]$ whenever $\eta_k \le 2\min_i p_i/L_i$. With the stricter choice $\eta_k \le \min_i p_i/L_i$, we obtain the uniform coefficient $1/2$ in Theorem 1. - Importance sampling improves constants: Choosing $p_i \propto L_i$ maximizes $\min_i p_i/L_i$ subject to $\sum_i p_i = 1$, yielding the largest allowable stepsize and best constants in the rate. - Uniform sampling baseline: If $p_i = 1/d$, then $\eta_k \le 1/(d\,L_{\max})$, where $L_{\max} := \max_i L_i$, and the rate constant scales with $d\,L_{\max}$. With $p_i \propto L_i$, one can take $\eta_k \le 1/\sum_i L_i$, improving dependence from $d\,L_{\max}$ to $\sum_i L_i$. - Hyperparameter sensitivity: The rate is inversely proportional to $\eta_{\min}$; picking η_k close to $\min_i p_i/L_i$ improves the constant. Too large η_k (exceeding $2\min_i p_i/L_i$) may destroy descent guarantees.

Discussion

The algorithm behaves like stochastic gradient descent with an unbiased gradient estimator that updates a single coordinate per iteration. The importance factor $1/p_{i_k}$ is crucial to make the estimator unbiased and to achieve the descent bound that aggregates all coordinates symmetrically through $\left\|\nabla f(w^k)\right\|^2 = \sum_i \nabla_i f(w^k)^2$. Coordinate-wise smoothness suffices to control the one-dimensional local quadratic approximation along the sampled coordinate, yielding the standard nonconvex stationarity bound $\mathcal{O}(1/T)$ for the expected gradient norm squared.

Preliminaries (Definitions and Lemmas)

We use the shorthand $\nabla f(w) = (\nabla_1 f(w), \dots, \nabla_d f(w))^\top$ and $\|\cdot\|$ for the Euclidean norm.

Lemma 1 (Univariate descent along a coordinate)**.** *Under coordinate-wise smoothness, for any $w \in \mathbb{R}^d$, coordinate i, and scalar $d \in \mathbb{R}$,*

$$f(w + d\,e_i) \;\le\; f(w) + \nabla_i f(w)\,d + \frac{L_i}{2}d^2.$$

Moreover, with $d = -\alpha\,\nabla_i f(w)$ for $\alpha \ge 0$,

$$f(w - \alpha\,\nabla_i f(w)\,e_i) \;\le\; f(w) - \alpha\Big(1 - \frac{L_i\alpha}{2}\Big)\,\nabla_i f(w)^2.$$

Proof. Let $\phi(t) := f(w + t\,e_i)$ for $t \in \mathbb{R}$. Then $\phi'(t) = \nabla_i f(w + t\,e_i)$ and by coordinate-wise L_i-Lipschitz continuity of $\nabla_i f$ we have $|\phi'(t) - \phi'(0)| \le L_i|t|$. Integrating from 0 to d,

$$\begin{aligned}\phi(d) - \phi(0) &= \int_0^d \phi'(t)\,dt = \int_0^d \big(\phi'(0) + (\phi'(t) - \phi'(0))\big)\,dt \\ &\le d\,\phi'(0) + \int_0^{|d|} L_i t\,dt = d\,\nabla_i f(w) + \frac{L_i}{2}d^2,\end{aligned}$$

which yields the first inequality. Substituting $d = -\alpha\,\nabla_i f(w)$ gives

$$f\big(w - \alpha\,\nabla_i f(w)\,e_i\big) \le f(w) - \alpha\,\nabla_i f(w)^2 + \frac{L_i}{2}\alpha^2\nabla_i f(w)^2 = f(w) - \alpha\Big(1 - \frac{L_i\alpha}{2}\Big)\nabla_i f(w)^2. \quad \square$$

Lemma 2 (One-step expected decrease)**.** *Let* $w^{k+1} = w^k - \eta_k \frac{1}{p_{i_k}} \nabla_{i_k} f(w^k)\, e_{i_k}$ *with* $i_k \sim p$ *independent of the past, and let* L_i *be coordinate-wise smoothness constants. Then*

$$\mathbb{E}\Big[f(w^{k+1}) \mid w^k\Big] \;\le\; f(w^k) - \eta_k \sum_{i=1}^{d} \left(1 - \frac{L_i \eta_k}{2p_i}\right) \nabla_i f(w^k)^2.$$

In particular, if $\eta_k \le \min_i p_i/L_i$*, then*

$$\mathbb{E}\Big[f(w^{k+1}) \mid w^k\Big] \;\le\; f(w^k) - \frac{\eta_k}{2} \left\|\nabla f(w^k)\right\|^2.$$

Proof. Condition on w^k. For the sampled i_k, set $\alpha_{k,i_k} := \eta_k/p_{i_k} \ge 0$. Apply Lemma 1 with $i = i_k$ and $\alpha = \alpha_{k,i_k}$:

$$\begin{aligned} f(w^{k+1}) &= f\Big(w^k - \alpha_{k,i_k}\, \nabla_{i_k} f(w^k)\, e_{i_k}\Big) \\ &\le f(w^k) - \alpha_{k,i_k}\Big(1 - \tfrac{L_{i_k}\alpha_{k,i_k}}{2}\Big) \nabla_{i_k} f(w^k)^2 \qquad \textbf{[Step 1]: by Lemma 1 with } d = -\alpha_{k,i_k} \nabla_{i_k} f(w^k). \end{aligned}$$

Take conditional expectation over $i_k \sim p$:

$$\begin{aligned} \mathbb{E}\Big[f(w^{k+1}) \mid w^k\Big] &\le f(w^k) - \sum_{i=1}^{d} p_i \left(\frac{\eta_k}{p_i}\right)\left(1 - \frac{L_i}{2}\frac{\eta_k}{p_i}\right) \nabla_i f(w^k)^2 \qquad \textbf{[Step 2]: linearity of } \mathbb{E} \text{ and substitution } \alpha \\ &= f(w^k) - \eta_k \sum_{i=1}^{d} \left(1 - \frac{L_i \eta_k}{2p_i}\right) \nabla_i f(w^k)^2 \qquad \textbf{[Step 3]: algebraic simplification}. \end{aligned}$$

If $\eta_k \le \min_i p_i/L_i$, then $0 \le \frac{L_i \eta_k}{2p_i} \le \frac{1}{2}$ for all i, hence $1 - \frac{L_i\eta_k}{2p_i} \ge \frac{1}{2}$. Therefore

$$\mathbb{E}\Big[f(w^{k+1}) \mid w^k\Big] \le f(w^k) - \frac{\eta_k}{2} \sum_{i=1}^{d} \nabla_i f(w^k)^2 = f(w^k) - \frac{\eta_k}{2} \left\|\nabla f(w^k)\right\|^2 \qquad \textbf{[Step 4]: use } 1 - \tfrac{L_i\eta_k}{2p_i} \ge \tfrac{1}{2} \text{ and } \sum_i \nabla_i^2$$

$\square$

Main Proof (Step-by-Step Derivation)

Proof of Theorem 1. Start from Lemma 2 and take total expectation using the tower property:

$$\begin{aligned} \mathbb{E}\big[f(w^{k+1})\big] &= \mathbb{E}\big[\,\mathbb{E}[f(w^{k+1}) \mid w^k]\,\big] \qquad \textbf{[Step 5]: tower property } \mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}]]. \\ &\le \mathbb{E}\big[f(w^k) - \tfrac{\eta_k}{2} \left\|\nabla f(w^k)\right\|^2\big] \qquad \textbf{[Step 6]: apply Lemma 2 with } \eta_k \le \min_i p_i/L_i. \\ &= \mathbb{E}[f(w^k)] - \frac{\eta_k}{2}\,\mathbb{E}\big[\left\|\nabla f(w^k)\right\|^2\big] \qquad \textbf{[Step 7]: linearity of expectation}. \end{aligned}$$

Rearrange:

$$\frac{\eta_k}{2}\,\mathbb{E}\big[\left\|\nabla f(w^k)\right\|^2\big] \;\le\; \mathbb{E}[f(w^k)] - \mathbb{E}[f(w^{k+1})] \qquad \textbf{[Step 8]: move terms to the right-hand side}.$$

Sum from $k = 0$ to $T - 1$:

$$\sum_{k=0}^{T-1} \frac{\eta_k}{2}\,\mathbb{E}\big[\,\big\|\nabla f(w^k)\big\|^2\,\big] \le \sum_{k=0}^{T-1}\Big(\mathbb{E}[f(w^k)] - \mathbb{E}[f(w^{k+1})]\Big) \qquad \textbf{[Step 9]: sum both sides over } k.$$

$$= \mathbb{E}[f(w^0)] - \mathbb{E}[f(w^T)] \qquad \textbf{[Step 10]: telescoping sum } \sum_{k=0}^{T-1}(a_k - a_{k+1}) = a_0 - a_T.$$

Use the lower bound $f(w^T) \ge f_\star$:

$$\sum_{k=0}^{T-1} \frac{\eta_k}{2}\,\mathbb{E}\big[\,\big\|\nabla f(w^k)\big\|^2\,\big] \;\le\; f(w^0) - f_\star \qquad \textbf{[Step 11]: apply Assumption 1}.$$

This proves the first claim.

For the random index R uniformly distributed in $\{0, \dots, T-1\}$ and independent of the history,

$$\begin{aligned}
\mathbb{E}\big[\,\|\nabla f(w^R)\|^2\,\big] &= \frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}\big[\,\big\|\nabla f(w^k)\big\|^2\,\big] && \textbf{[Step 12]: law of total expectation over discrete uniform } R. \\
&\le \frac{2}{T}\sum_{k=0}^{T-1}\frac{1}{\eta_k}\left(\frac{\eta_k}{2}\mathbb{E}\big[\,\big\|\nabla f(w^k)\big\|^2\,\big]\right) && \textbf{[Step 13]: multiply and divide by } \eta_k > 0. \\
&\le \frac{2}{T}\cdot\frac{1}{\eta_{\min}}\sum_{k=0}^{T-1}\left(\frac{\eta_k}{2}\mathbb{E}\big[\,\big\|\nabla f(w^k)\big\|^2\,\big]\right) && \textbf{[Step 14]: since } \eta_k \ge \eta_{\min}. \\
&\le \frac{2}{T}\cdot\frac{1}{\eta_{\min}}\,(f(w^0) - f_\star) && \textbf{[Step 15]: apply the first claim}.
\end{aligned}$$

This concludes the proof. □

Rate Derivation (With Constants)

- Uniform sampling $p_i = \frac{1}{d}$: - Stepsize constraint: $\eta \le \min_i \frac{1/d}{L_i} = \frac{1}{d\,L_{\max}}$ with $L_{\max} = \max_i L_i$. - Rate: for $\eta = \frac{1}{d\,L_{\max}}$,

$$\mathbb{E}\big[\,\|\nabla f(w^R)\|^2\,\big] \le \frac{2d\,L_{\max}\,(f(w^0) - f_\star)}{T}.$$

- Importance sampling $p_i = \frac{L_i}{\sum_{j=1}^d L_j}$: - Stepsize constraint: $\eta \le \min_i \frac{L_i/\sum_j L_j}{L_i} = \frac{1}{\sum_j L_j}$. - Rate: for $\eta = \frac{1}{\sum_j L_j}$,

$$\mathbb{E}\big[\,\|\nabla f(w^R)\|^2\,\big] \le \frac{2\Big(\sum_{j=1}^d L_j\Big)\,(f(w^0) - f_\star)}{T}.$$

In general, the constant scales like $1/\eta_{\min}$, and larger $\min_i p_i/L_i$ allows larger η and better constants.

Special Cases

- One dimension ($d = 1$): $p_1 = 1$, the update reduces to gradient descent with stepsize η, and Theorem 1 recovers the standard nonconvex descent bound for L_1-smooth functions with $\eta \le 1/L_1$. - Equal smoothness ($L_i \equiv L$): uniform sampling yields $\eta \le 1/(dL)$ and rate constant $2dL$; importance sampling coincides with uniform in this case.